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The Palaeomagnetism of the Ethiopian Flood Basalt Succession near Addis Ababa

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Summary

Measurements on samples from 20 sites in the Ethiopian flood basalt succession yield a palaeomagnetic pole at 80·8° N, 167·7° E with a circle of confidence of 4·3°. The pole for these rocks is significantly different from the present geographic pole, and is in general agreement with other results from Miocene formations in Africa. The results jointly suggest that the palaeomagnetic pole for Africa was displaced from the present pole by about 6° for most of the mid-Tertiary (Eocene to Miocene).

1. Introduction

The study of the later stages of the fragmentation of Gondwanaland, and in particular the post-Cretaceous movement of Africa, is hampered by the lack of palaeomagnetic data from early and mid-Tertiary rocks. In the Cretaceous the African palaeomagnetic pole was about 25° from the late Tertiary pole which itself coincided closely with the present geographic pole (McElhinny, *et al.* 1968). In order to study the nature of the polar shift producing this change it is necessary to sample rocks from older Tertiary formations. Unfortunately the Tertiary volcanic rocks from East Africa are generally Miocene or younger. However, to the north in Ethiopia there are thick sequences of Eocene–Oligocene flood basalt lavas.

The Ethiopian lavas were sampled briefly by Grasty (1964) who obtained 21 samples from 11 sites in the basalt succession exposed in the Blue Nile Gorge north of Addis Ababa. The polarity of all samples was normal, and the palaeomagnetic pole is at 87° N, 253½° E, with a circle of confidence of 10½°. The small sample and the large error do not allow one to decide whether the pole differs significantly from the present geographic pole and clearly further work was justified. Accordingly, two of us (A.B. and P.G.) in December 1967 carried out sampling in the deep gorges cutting the flood basalt succession near Addis Ababa, studying sections selected by the third author (I.L.G.) on the basis of a preliminary reconnaissance of the area.

2. Geology

The formation of the Rift System in the Ethio-Arabian area was associated with the eruption of thick sequences of flood basalts. These lavas, commonly known as the Ethiopian and Yemen Trap series, usually overlie Mesozoic sediments. The

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basalts outcrop principally on the Ethiopian and Somali Plateaux and in the Yemen highlands. The lava sequences are generally thickest in areas adjacent to the rifts and thin as they are traced away from these possible source regions. The geology of these lava piles is poorly known. This is particularly the case in Ethiopia where the thickness of the succession may exceed 3000 m, and where there are very considerable structural complications at the margins of the rifts. Mohr (1968) has sub-divided the Ethiopian basalt succession into a lower Trap Series and an upper, overlying Shield Group. However, we prefer to consider the basalt sequence as a single, complex stratigraphic unit.

The age of the Ethiopian flood basalt succession is poorly known, but probably ranges from Eocene to Miocene. The sequence is almost certainly not everywhere of the same age and intra-basalt unconformities occur (Mohr & Rogers 1966). In view of these geological difficulties, the sections selected for palaeomagnetic work were chosen with care. The three sampled sections are on the Ethiopian Plateau in areas away from the rift margins. The basalts here form a flat-lying unfaulted succession. Although the succession is rather thin in these areas the flows are unaltered or only weakly zeolitized. As the stratigraphy of the succession is unknown, the sections studied were purposely chosen from a limited area in the hope of providing a concordant group of results as a basis for later work. The locations are shown on the map (Fig. 1) and brief notes on each section are given below.

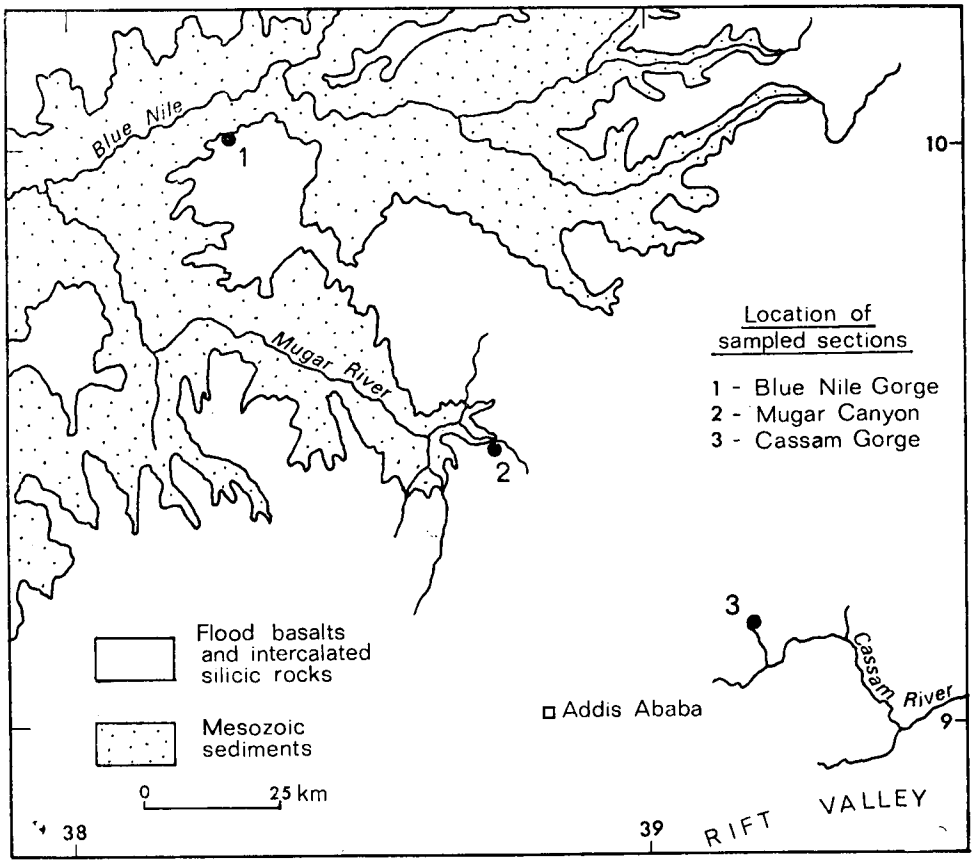


FIG. 1. Geological sketch-map of the Ethiopian Plateau showing the location of the sampled sections.

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Blue Nile Gorge section

The most northerly section studied is on the southern side of the Blue Nile Gorge, just to the north of the village of Tulu Milki (Lat. 9° 55′ N, Long. 38° 21′ E). The basalt lavas here form a sequence approximately 220 m thick and the flows were sampled along the main Addis Ababa–Gondar Road. The section thus corresponds closely to that studied by Grasty (1964) although subsequent re-alignment of the road has made it difficult to re-sample the identical section.

The basalt sequence rests with apparent conformity on a 100 m thick crystalline limestone tentatively assigned to the Upper Jurassic. However, the absence of the upper regressive sandstone *facies* from the top of the Mesozoic succession suggests that erosion and hence a considerable period of time might have elapsed prior to the eruption of the lowermost basalt lava. Only five discrete lava flows were identified with certainty, and this thin sequence is broken by at least two horizons of intercalated volcano-clastic sediments, one of which is at least 30 m thick. The presence of these intra-basaltic horizons might be taken to indicate that the basalts were extruded over a considerable period of time, perhaps in excess of two million years.

The age of the basalts is poorly defined by the stratigraphic evidence cited above. However, the samples studied by Grasty (1964) were later dated by K/A measurements (Grasty, Miller & Mohr 1963). Five samples gave ages between 30 and 69 My with an average apparent age of 49 My $\pm$ 15 My. Thus the Blue Nile Gorge section is tentatively assigned to the Eocene–Oligocene.

Mugar Canyon section

This section is situated some 50 km north of Addis Ababa at the head of the gorge cut by the Mugar River at approximately Lat. 9° 25′ N, Long. 38° 45′ E. The lavas here outcrop in a steep cliff and the samples were collected down the north-east face of the cliff along a track which descends from a point near the top of the Mugar Canyon Aerial Ropeway.

The samples were collected from the topmost eleven flows, the total thickness of the sequence exceeding 200 m. Unfortunately time did not permit the sampling of the bottom two or three flows. The basalts rest directly on a thick red sandstone unit which here forms the top of the Mesozoic succession. Relations are thus comparable to those observed at the Blue Nile section, some 75 km to the north-west, and the two sections are thought to be approximately stratigraphically equivalent. However, the greater thickness of the Mugar Caynon section and the absence here of intercalated volcano-clastic horizons indicate that this section is closer to the source vents for the lavas which are probably located along the present margin of the Ethiopian Rift.

Cassam Gorge

The third studied section is located about 50 km east-north-east of Addis Ababa, close to the south-eastern margin of the Ethiopian Plateau (Lat. 9° 10′ N, Long. 39° 15′ E). The succession studied is more complicated than in either of the previous two sections and is given in detail in Fig. 2. It should be noted that the Mesozoic sediments are not exposed at this point and that the volcanic succession has a thickness of at least 250 m again emphasizing the thickening of the basalt succession as it is traced towards the Rift Valley margin. The topmost unit studied is a 20 m thick silicic ignimbrite which here lies with complete conformity on the underlying basalt lavas. It has therefore been treated, for palaeomagnetic purposes, as part of the lava succession. However, recently Mohr (1968) has suggested that this and other similar units might be Pliocene in age. If this is indeed the case, there was either a

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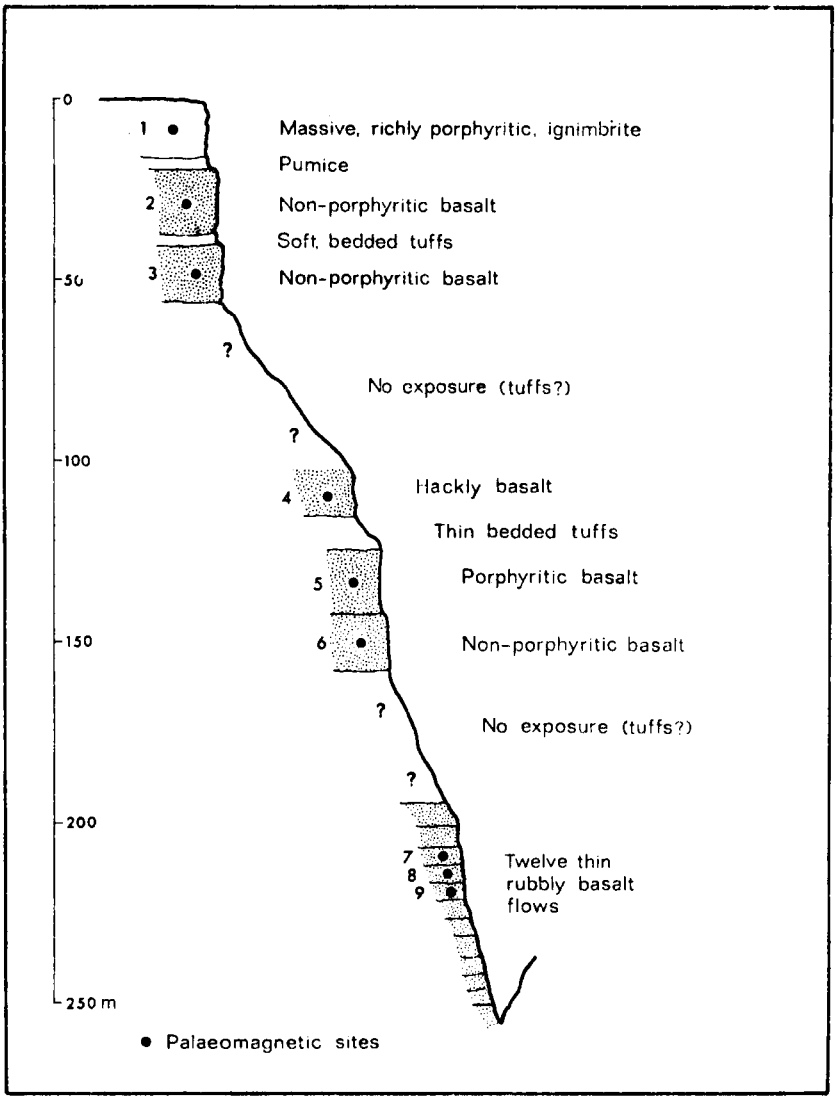


FIG. 2. The geological section studied at the Cassam Gorge showing the palaeomagnetic sites.

considerable time interval between the eruption of the basalts and the topmost ignimbrite, or perhaps more likely, the basalts exposed in this section are younger than those sampled at the Blue Nile and Mugar Canyon sections.

3. Sampling and measurement

Samples were collected by the normal block method, after orientation with magnetic compass and clinometer. Each independently oriented block was regarded as a sample, and two or more samples were collected from each flow. At the Mugar Canyon section 31 samples were collected from 11 flows, at the Blue Nile Gorge 16 samples from five flows, and at the Cassam Gorge 18 samples were taken from nine flows. A total of 65 blocks were therefore collected from 25 flows.

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Several cores 2·2 cm in diameter were taken from each block using an electric press drill especially adapted for diamond coring in the engineering workshop at Haile Sellassie I University, Addis Ababa. At the laboratory in Nairobi cylindrical specimens 2·5 cm in length were cut from the cores. Thus each sample provided two or more specimens. The intensity and direction of magnetization of specimens was measured under an astatic magnetometer with a maximum sensitivity of $1{\cdot}5 \times 10^{-5}$ e.m.u. of moment per mm of scale deflection. All samples were magnetically ' cleaned ' by subjecting them to partial demagnetization in alternating fields ranging from 100 to 600 Gauss. The demagnetizing field for all specimens from a flow was selected by subjecting one specimen to a stepwise demagnetization at intervals of 50 Gauss. In most cases soft secondary components of magnetization were removed in fields between 50 and 100 Gauss.

In the analysis, samples rather than specimens were regarded as the primary units of data. Specimens from a single sample are always close to one-another, and are subject to a common orienting error. The separate specimen directions were used to check for agreement within a single block. When reasonable agreement was obtained, a mean of the specimen directions was taken, and then used as the sample direction. In practice, the main variability was between samples rather than between specimens within a sample. The procedure described here was applied both to the initial measurements and to the measurements after partial demagnetization.

4. Results

Between two and five samples, defined as above, were taken from each flow. Following normal palaeomagnetic practice each flow was regarded as a separate site, and site mean directions were calculated from all samples within each flow. Table 1 shows the site directions before and after demagnetization, and the corresponding palaeomagnetic pole positions. A number of samples (four in total) were lost during transport from the field. Five others showed erratic behaviour during demagnetization yielding no stable end-point, and could not be included in the analysis. One further factor leading to exclusion was the adoption of a rule that no site should be represented in the final analysis by only one sample. The argument here is similar to that used earlier in selecting satisfactory samples; it is based upon the need for independent confirmation from at least two separate sets of measurements. The application of this rule eliminated two sites from the final analysis.

For all these reasons the numbers in Table 1 show slight variations from those given in the account of the sampling procedure. In outline, 65 samples from 25 sites were collected in the field. The initial measurements involve 61 samples from 25 sites, and after magnetic cleaning 54 samples from 21 sites give good stable end points. The mean directions of these 21 sites after cleaning, are believed to be good estimates of the thermo-remanence acquired at the sites at the time of cooling.

Two of the 21 sites require particular mention. Site No. 5 in the Cassam Gorge gave a stable direction which was about 100° from the average direction of the remainder. Stability was demonstrated by demagnetization in fields up to 400 Gauss, and there is little doubt that the site direction is representative of the original thermo-remanence. The direction is therefore assumed to be an ' Intermediate ' direction such as sometimes occur in sequences of lava flows (Dagley, *et al.* 1967). It is usually assumed that such directions may occur during the process of a reversal. In this particular case the flows both above and below are normally polarized, although the overlying flow (No. 4) has a direction which is about 20° from the mean. As will be explained below a deviation from the mean of 20° does not exclude a site from the group, but it is, perhaps, an indication that the Earth's field was unusually disturbed at the time of formation of this flow. A short reversed event cannot therefore be ruled

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Table 1

Site data before and after demagnetization

Section		Initial remanence			After demagnetization				Palaeomagnetic North Pole	
Site/Flow	*N*	*D*°	*I*°	*k*	*N*	*D*°	*I*°	*k*	Lat. °N	Long. °E
Mugar Canyon										
1	4	357·7	27·6	7·6	4	358·5	8·8	33·5	84·7	235·5
2	2	303·8	−23·6	1·9		Unstable				
3	3	347·7	−17·7	5·3	3	357·9	8·3	27·1	76·2	227·3
4	5	358·2	9·4	14·4	5	1·4	0·9	18·3	80·9	210·0
5	2	18·0	11·0	340	2	17·0	7·0	88·9	72·1	146·9
6	2	8·0	9·5	15·6	2	6·6	5·5	8·3	80·6	173·9
7	2	17·0	5·0	660	2	15·3	7·3	183	73·8	148·6
8	3	13·3	− 2·4	84·7	3	13·7	− 6·4	62·5	71·4	170·9
9	2	3·2	−16·6	50·5	2	11·6	−10·6	36·6	71·2	180·1
10	2	4·8	18·3	9·9	2	356·0	21·3	15·7	85·8	330·3
11	2	0·1	15·0	48·6	2	359·0	16·9	12·3	88·7	268·2
Blue Nile Gorge										
1	3	343·3	30·2	4·1	3	359·1	19·8	12·8	89·1	320·7
2	3	348·9	9·8	49·1	3	357·8	11·4	111	85·2	245·3
3	3	20·3	17·0	128	3	20·5	16·0	151	69·7	131·8
4	4	8·1	16·4	34·0	4	4·4	16·1	13·2	85·3	150·7
5	1	9·0	−25·0			One sample only				
Cassam Gorge										
1	2	350·7	7·7	14·6	2	8·5	−11·5	101	72·8	189·7
2	2	11·5	8·1	36·8	2	359·3	4·5	98·7	83·0	225·5
3	2	10·5	7·5	1330	2	13·0	2·0	85·9	74·7	161·0
4	2	23·4	22·6	5·8	2	30·7	13·2	32·4	59·5	131·9
5	2	345·6	−27·2	15·6	2	272·0	−65·0	250	Intermediate Direction	
6	2	11·6	23·6	29·5	2	10·1	20·9	55·8	79·9	119·2
7	2	8·0	4·1	22·4		Unstable				
8	2	24·4	− 6·0	26·2		One sample only				
9	2	181·5	− 7·0	77·1	2	183·5	− 7·6	44·9	83·6° S	6·6° E

N, Number of samples; *D*, declination in degrees east of true north; *I*, inclination in degrees measured positive downwards; and *k*, Fisher's precision parameter.

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out, especially when it is remembered that there is no clear evidence regarding the time interval between flows in this part of the section.

Site 9 from the lowest sampled lava in the Cassam Gorge (see Fig. 2) is the only reversed site. The reversal is already suggested by the reversed polarity of the initial remanence, and is confirmed by demagnetization in alternating fields up to 400 Gauss. The two overlying flows are excluded from the final analysis, but it is indicative that the only stable sample from these two flows was also reversed, suggesting that all three flows might have reversed polarity. Our results, however, can only confirm reversed polarity for Site 9.

5. Analysis

Table 2 shows the mean values for each section giving unit weight to sites. Before computing the mean for the Cassam Gorge section, three sites required special consideration. Site 5 which we have already seen to have an intermediate direction was excluded from the mean since it differs from the mean by more than 100°. Site 4 has a direction which is about 20° from the mean. Using the approximate expression, $\theta_{95} = 140/\sqrt{k}$, for the radius of the circle which contains 95 per cent of the directions (Irving 1964, p. 60), we find for the Cassam Gorge, $\theta_{95} = 26{\cdot}4°$, and there is therefore no reason to believe that Site 4 is not a sample from the Cassam Gorge population. Hence it is included in the mean.

Site 9 has been confirmed as having reversed polarity. Since it is the only reversed site, is is not possible to test for the identity of the reversed and normal groups. But when the direction for Site 9 is reversed in sign it is found to be only 7° from the mean. Accordingly this site is included in the mean with its added reversal.

The mean directions for the three sections together with their circles of confidence are shown in Fig. 3. The directions and circles of confidence of the sections overlap suggesting that the three sets of results may be combined. An overall mean for the 20 sites was therefore calculated, and appears both in Table 2 and in Fig. 3. This mean direction, and the corresponding pole, calculated from the individual site poles, are the principal outcome of this study. It is believed that they are good estimates of the mean field direction and palaeomagnetic pole for the sampled part of the Ethiopian flood basalt succession.

It will be seen that the pole is about 9° from the present geographic pole, and with a circle of confidence of just over 4° this 9° polar difference becomes significant. Of the three sections sampled, only the Blue Nile Gorge failed to show a similar difference and this can be understood in view of its larger circle of confidence arising from a smaller number of sites. The mean value for the 20 sites therefore allows us to state that on the evidence presented here, and making the usual geocentric axial dipole assumption, the palaeomagnetic pole for Africa at the time of formation of the basalt succession was at 80·8° N, 167·7° E with a circle of confidence of 4·3°.

The implications of this result will be discussed later. Meanwhile it is of some interest to compute the result in an alternative way. The data from the 20 sites and 52 samples was subjected to a two-tier analysis as described by Watson & Irving (1957). It yields as estimates of the between-site and within-site precision parameters, the following values: $K_b = 86{\cdot}4$, $K_w = 25{\cdot}8$. The estimate of the mean direction is given in a two-tier analysis by the overall sample mean. Table 3 shows this to be $D = 6{\cdot}3°$, $I = +7{\cdot}2°$, and the circle of confidence for between-site variation has a value of 5·1°, slightly smaller than the α_{95}, shown in Table 2. The difference between the final mean calculated by the simpler method utilizing site means, and that obtained from the two-tier analysis is less than 1° and is unimportant for the final result.

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Table 2

Means giving sites unit weight

	Initial remanence					After demagnetization					Palaeomagnetic North Pole			
Section	N	$D°$	$I°$	k	$\alpha_{95}°$	N	$D°$	$I°$	k	$\alpha_{95}°$	Lat. °N	Long. °E	K	$A_{95}°$
Mugar Canyon	11	0·4	3·5	10·4	14·9	10	5·8	4·3	37·8	8·0	80·8	180·3	70·1	5·8
Blue Nile Gorge	5	2·1	10·3	10·4	24·9	4	5·4	16·0	58·8	12·1	84·4	147·8	60·2	11·9
Cassam Gorge	9	9·5	2·7	3·2	34·1	6	10·7	6·2	28·0	12·9	77·7	158·1	44·3	10·2
Overall mean after cleaning						20	7·2	7·2	34·9	5·6	80·8	167·7	59·3	4·3

N, Number of sites; D, I, Declination and inclination as in Table 1; k, K, Fisher's precision parameter estimate; and α_{95}, A_{95} radius of circle of 95 per cent confidence.

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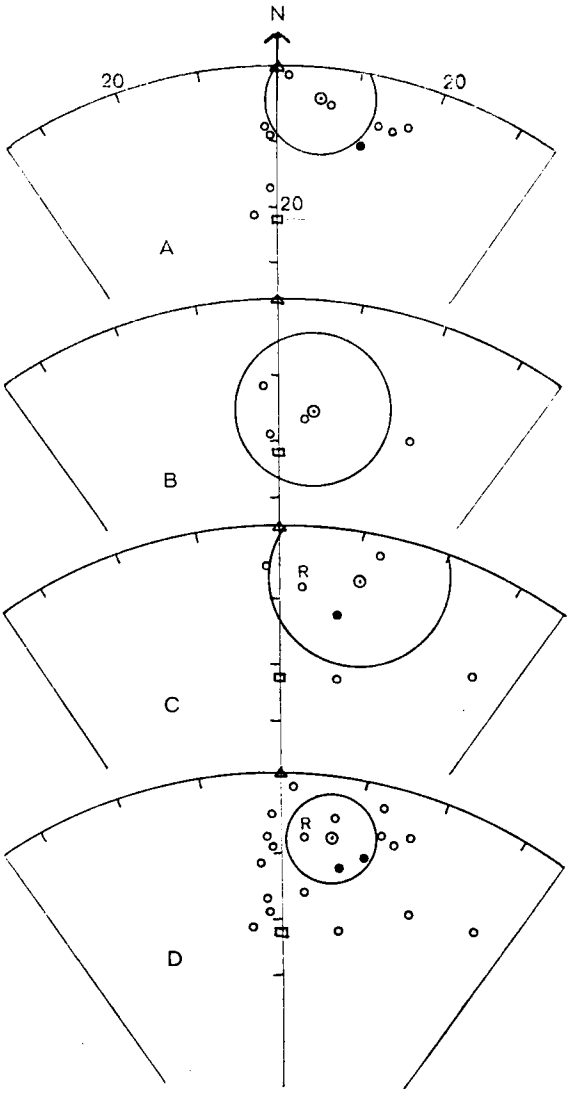


FIG. 3. Stereographic projections showing cleaned site directions and mean directions. A. Mugar Canyon section. B. Blue Nile Gorge section. C. Cassam Gorge section. The one reversed sited is shown as a south seeking vector and is marked R. The ' intermediate ' site is not shown. D. Combination of the three sections showing all sites ($N = 20$) and their mean. Open symbols, lower hemisphere; solid symbols, upper hemisphere; mean direction is indicated by an open circle and dot at the centre of a circle showing the 95 per cent confidence limit (α_{95}); open triangle, present field at Addis Ababa; open square, dipole field at Addis Ababa.

Table 3

Means giving samples unit weight after demagnetization

Section	N	$D°$	$I°$	k	$\alpha_{95}°$
Mugar Canyon	27	4·8	3·3	20·4	6·3
Blue Nile Gorge	13	5·4	16·0	21·2	9·2
Cassam Gorge	12	10·7	6·2	24·1	9·0
Overall	52	6·3	7·2	20·0	4·6

N, Number of samples; other symbols as in Table 2.

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6. Discussion

The only other palaeomagnetic study of the Ethiopian flood basalts is that due to Grasty (1964). His results give a pole at 87° N, 253½° E with a circle of confidence of 10½°. With its larger error it is not significantly different from the result of this study, and there is nothing to suggest that the two results are incompatable. However, the larger error means that Grasty's result is not able to show the 9° polar difference which this more detailed study has revealed. The presence of a small polar difference raises the question whether a similar difference can be seen in any other African result.

In a very detailed study of the Miocene Turkana lavas which outcrop in N.W. Kenya, Raja (1968) has found a significant polar difference of about 5°. The Turkana pole is at 84·8° N, 163·3° E with a circle of confidence of 2·3°. This pole is only 4° from the Ethiopian pole: the circles of confidence of the two poles overlap, and neither includes the present geographic pole. The two results therefore support one another. Further support is available from unpublished results by Mr T. A. Reilly (personal

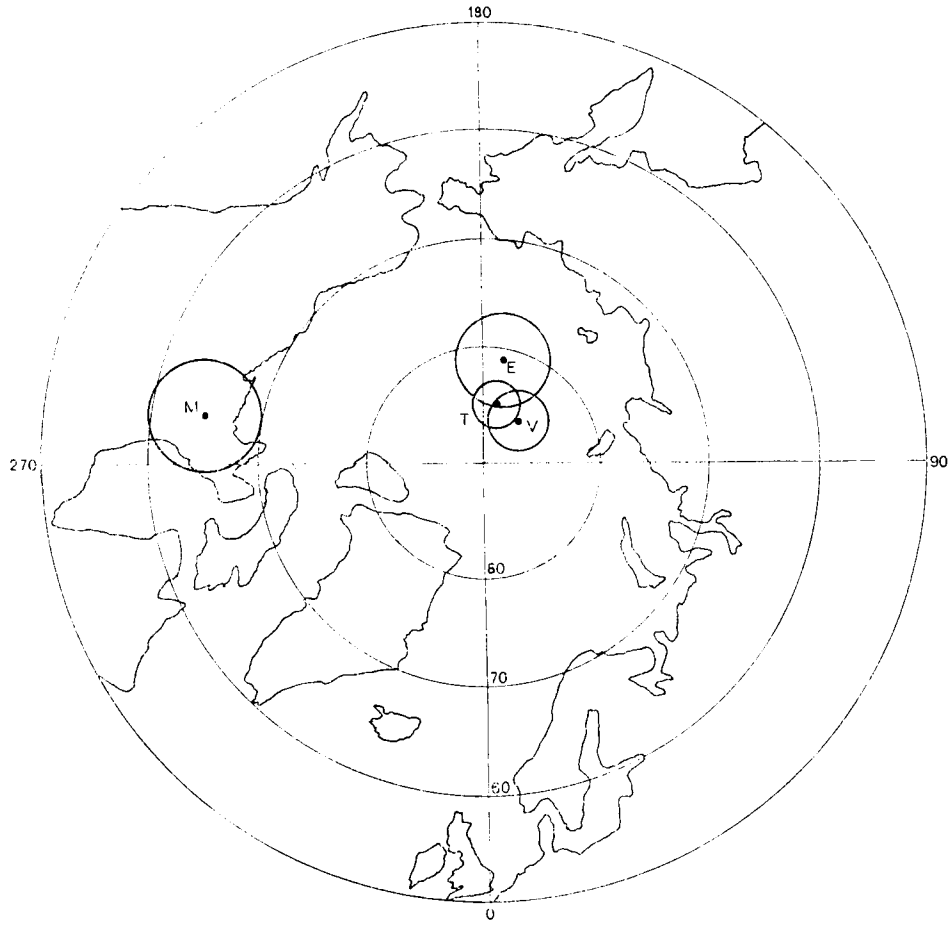


FIG. 4. Equal area projection showing palaeomagnetic poles and their circles of confidence (A_{95}). E. Ethiopian flood basalts, this study; T. Turkana lavas (Raja 1968); V. Cape Verde Islands (Watkins, *et al.* 1968); and M. ' Mean Mesozoic pole ' (McElhinny, *et al.* 1968).

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communication). He has found that the pole for rocks of Lower Pliocene–Miocene age in Kenya also differs significantly from the present geographic pole. But the younger rocks studied by Mr Reilly do not seem to show this: they suggest that significant polar movement has not occurred in the past 7 My. The movement which produced the effects shown by the older East African lavas and the Ethiopian Traps must therefore have occurred at a time greater than 7 My, and must have been almost complete at about 10–11 My near the Pliocene–Miocene boundary.

Another Miocene result has recently been published by Watkins, Richardson & Mason (1968) who find a palaeomagnetic pole for lavas of the Cape Verde Islands at 85·2° N, 136·6° E with an oval of confidence having semi-axes 3·7° and 2·0°. This pole is very close to those already mentioned, and like them is just different from the geographic pole. The three poles are shown in Fig. 4. The close agreement of the Cape Verde Islands pole and the African poles strongly suggests that there has been no relative movement between the Cape Verde Islands and the African mainland since the Miocene. This would be in accord with current ideas that each continent and its adjacent sea-floor areas form a rigid plate (Le Pichon 1968).

Taken together, these results suggest a polar wander path for Africa which must show a small 5° deviation from the present pole in the Miocene, continuing and possibly enlarging to about 9° at the time of formation of the Ethiopian Traps. Previous results from the Tertiary of Africa have not been sufficiently detailed to show this effect (Brock 1968), although it has been suggested on other grounds (Girdler 1968).

Going further back in time the next reliable result for Africa is a Lower Cretaceous pole with an age of about 110 My (Briden 1967). It is one of a group of nine Mesozoic poles which McElhinny *et al.* (1968) have shown to be closely grouped to give a ' mean Mesozoic pole ' at about 65° N, 261° E. It is clear that between the youngest of the Mesozoic poles, and the oldest of the Tertiary results, there has been a large movement. A better age for the Ethiopian flood basalts would help to define more accurately the time interval during which the movement must have occurred. But for the nature of the movement itself, more results are required especially from the Lower Tertiary and the Upper Cretaceous.

This discussion has been framed in terms of the polar wander path. It could alternatively have been cast in terms of continental movements. For Africa, the Tertiary results show very small palaeolatitude changes, since polar movement is roughly perpendicular to the 40° E meridian. But by the same token, changes of palaeo-azimuth are relatively important. A Miocene palaeo-azimuth of about 5° (E of N) is required by the Turkana results, and of about 7° by the Ethiopian basalts. This implies a clockwise rotation of 6° for the movement of Africa into its present position since the Miocene. This may be compared with an anti-clockwise rotation of 7° suggested for the Arabian peninsular by palaeomagnetic studies of the Aden volcanics (Irving & Tarling 1961; Tarling, Sanver & Hutchings 1967). The relation between these two rotations depends upon their timing. Tarling, *et al.* (1967) give some rather scattered dates for the Aden Volcanics which do, however, suggest that the Arabian rotation occurred later than the African one, and probably took place entirely after Africa had reached its present position.

Palaeomagnetic results from Africa alone cannot distinguish between continental movement and polar wandering. If, however, results from other continents are available the possibility emerges of separating polar wander from continental drift. A recent detailed study of the Australian Tertiary (Wellman, McElhinny & McDougall 1969) has yielded a new polar wander path for Australia. In a subsequent publication (McElhinny & Wellman 1969) the Australian workers have used sea-floor spreading data to distinguish that part of the apparent polar wander path which is due to polar wander alone from that part which is due to continental drift. The ' true ' polar wander curve which they present shows an excursion from the present N pole which

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occurs roughly along the 180° meridian, and which reaches a maximum of about 13° at a time 34 My ago. The excursion shown by the African results reaches a value of about 9° in the Eocene–Oligocene, and occurs along a meridian of about 160° E. The two paths are similar and may represent an episode of true polar wander common to both continents. If this is the case, the apparent polar wander path for Africa from the mid-Tertiary to the present must be due almost entirely to polar wander alone, and a late-Tertiary rotation of Africa must be ruled out.

To return to the results given in this paper, and to the similar results from Turkana and from the Cape Verde Islands, it is necessary to make one cautionary remark. As stated earlier it is believed that the results demonstrate a small but real polar difference for the mid-Tertiary. The conclusion is correct, provided that the normal palaeomagnetic assumptions are themselves correct. The difficulty is that when relatively small differences ($\sim 5°$) are being sought, their confirmation depends critically upon the exactness of the palaeomagnetic assumptions. If the populations depart only slightly from a Fisherian form, or if the dipole is not quite axial, or geocentric, then incorrect conclusions may easily be drawn. The problem is less severe when angular differences are larger. The larger difference ($\sim 25°$) between the present pole and the Mesozoic pole is many times the expected error, and could occur only as a result of gross departures from the normal assumptions. The 5° difference shown by the Miocene pole is much smaller: one is working very much closer to the limits of the method. But the conclusions described here are formally correct, and until there is positive evidence that the palaeomagnetic assumptions do not hold, we cannot dismiss them.

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